

Chapter 7 — Mitigating bias in datasets

Once a disparity and a fairness objective are defined

Learning objectives

- Select an intervention using evidence rather than convenience

Pre-processing through reweighting

- Reweighting changes how strongly training instances influence learning
- Underrepresented groups can receive greater weight so that the majority group does not
- This approach preserves records while changing their contribution

Pre-processing through feature treatment

- Sensitive or strongly biased features may be removed
- Simple removal is not a guarantee of fairness because names, neighborhoods, schools
- Feature treatment should be followed by proxy analysis and subgroup evaluation

In-processing with fairness constraints

- Fairness-constrained learning modifies the training objective by adding a constraint or
- The model then optimizes predictive performance and the selected fairness criterion
- Quality of outcome labels

In-processing with adversarial debiasing

- Adversarial debiasing trains a predictor alongside an adversary that tries to infer a
- The predictor is penalized when the adversary succeeds
- The method can reduce proxy leakage

Post-processing with equalized odds

- Equalized-odds post-processing adjusts decisions or thresholds from an already trained
- Operational review
- They can also change accuracy and may become unstable when group samples are small

Post-processing with calibration

- Calibration adjusts predicted probabilities so that a score has a consistent empirical
- It is valuable when scores guide human decisions or resource allocation
- Calibration does not guarantee equal selection or error rates

Choosing and validating an intervention

- Intervention begins with a diagnosed mechanism and a stated fairness target
- Monitor drift after deployment
- Mitigation is an iterative control process rather than a one-time repair

Takeaways

- Pre-processing changes data, in-processing changes learning
- Output adjustment targets decision behavior or probability meaning
- Every method must be validated against the intended fairness objective

Next

- Complete the quiz for this part
- The next clip moves from individual techniques to operating practice